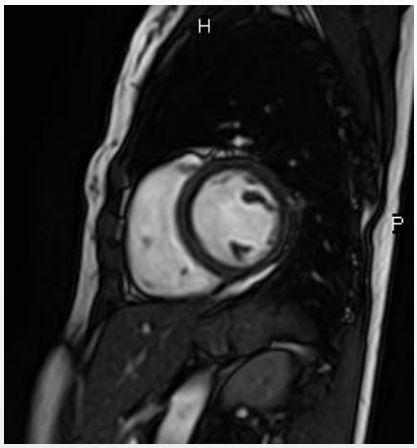
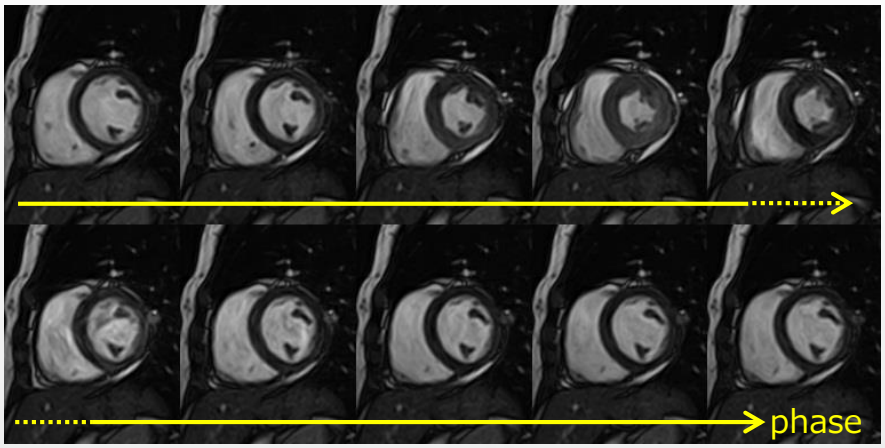
＊“mCardiac package” is required.

Cardiac MR : Cine MR

●Introduction

This is an imaging method to obtain continuous cardiac phase images from systole to diastole. Ventricular function and local wall motion can be evaluated using cardiac function analysis.



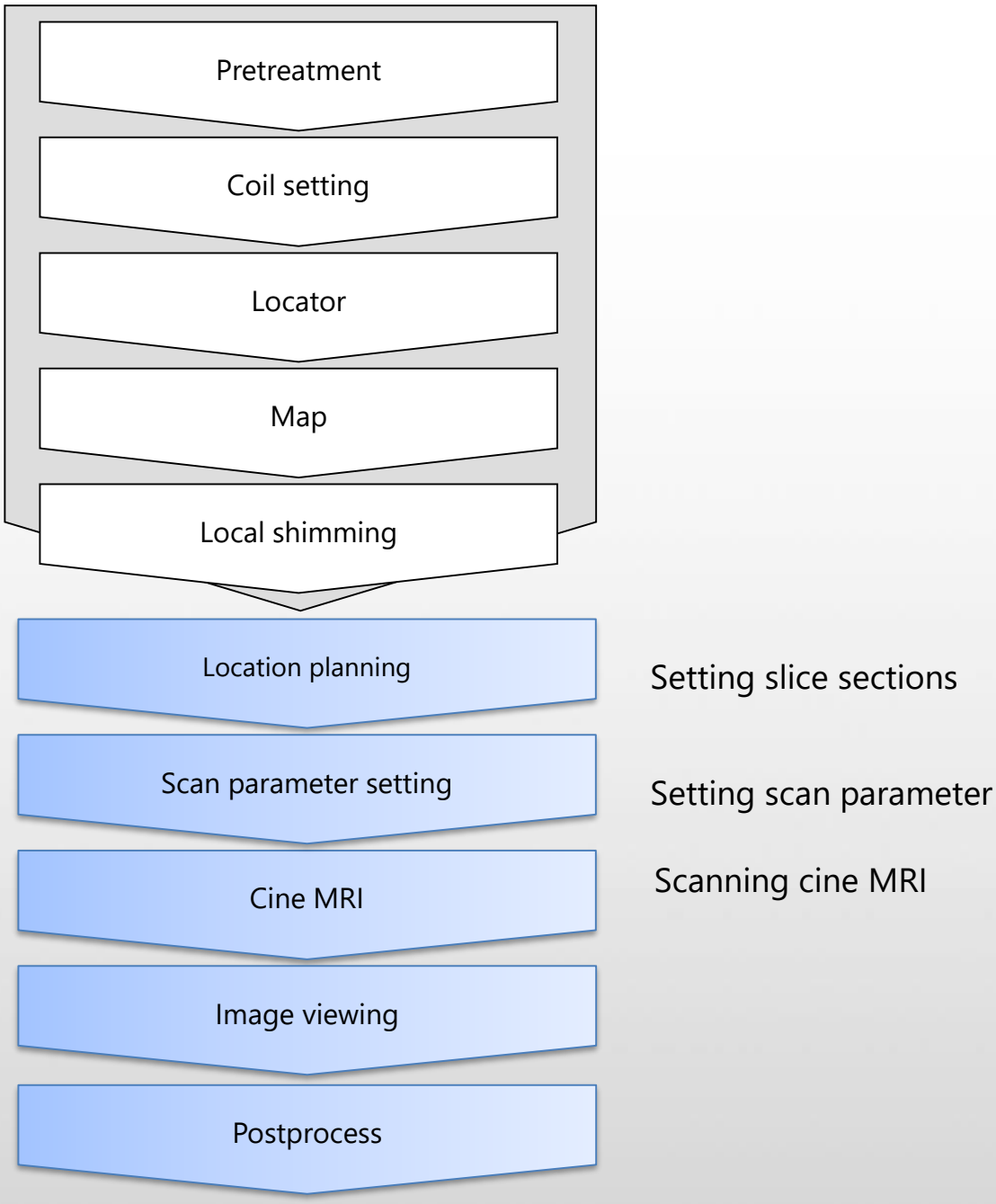


Cine MRI（LV short）

●imaging conditions

・SSFP is the method of choice for cine imaging as it provides high SNR and excellent contrast between myocardium and blood pool.
・All images are taken with ECG and also breath-hold.

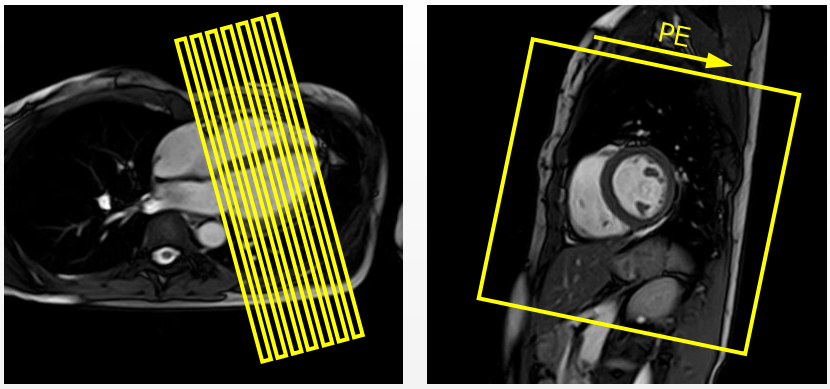
scanning workflow



●scanning workflow

Location planning Setting scan plan

Set the ROI to the desired imaging cross section (2-chamber long axis, horizontal long axis, left ventricular short axis, four-chamber long axis, three-chamber long axis, etc.) and adjust to the required number of slices. Rotate the ROI so there is no wrapping and check the encoding direction.



Scan planning of LV short axis

Scan parameter setting Set the imaging parameter and adjust the breath holding imaging time.

(1) Set the gate conditions and check the imaging conditions.

・Read the patient's R-R interval / heart rate.

・Set the maximum number of phases.

The maximum value that can be set can be adjusted depending on the PE-matrix and the number of segments.

・Set the flip angle to the maximum.

The deeper the flip angle, the higher the signal of blood and the lower signal of myocardium, so the contrast will be better.

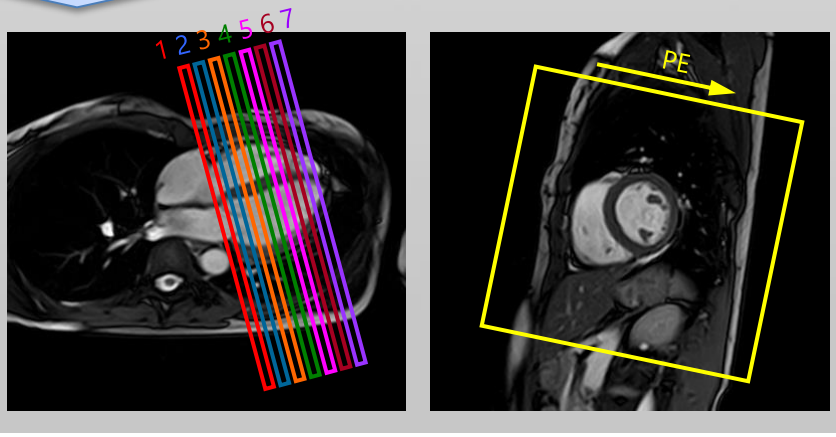
・Check the imaging time.

If you want to shorten the imaging time (breath holding time), adjust the PE-matrix and the number of segments.

(2) Explain breath-hold imaging to the patient.

・Since breath-hold imaging is repeated multiple times, ask the patient to hold your breath at the same position each time.

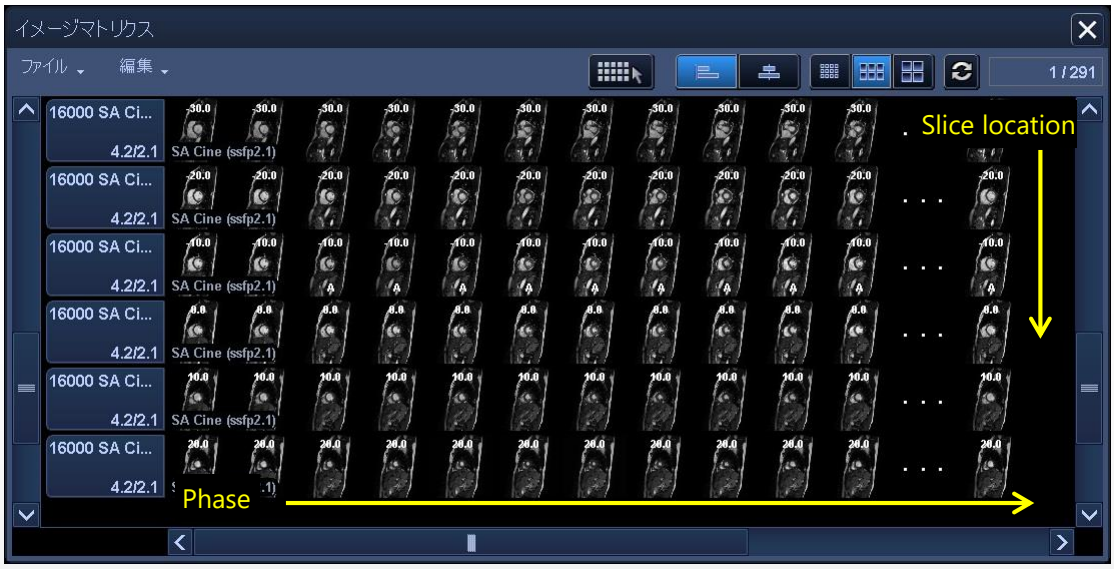
Cine MRI Scanning cine MRI



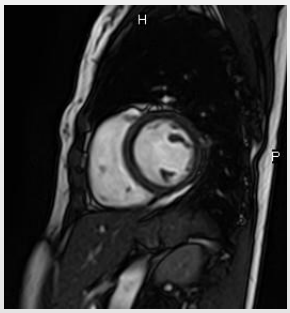
Scan planning of LV short axis

Image viewing

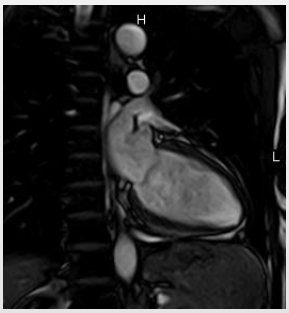
After image scanning, the image is displayed in the image matrix as shown below.



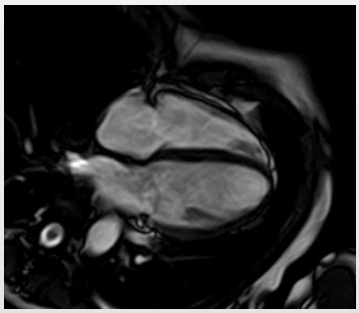
＊Cine can be displayed from [Cine display] on the Screening tab.



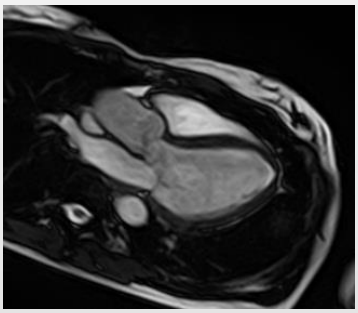
LV short



2ch long（vertical long）



4ch long



3ch long

Post process

Perform cardiac function analysis as needed.

※The Cardiac Analysis application (optional software) is required for analysis on the device itself. For details of processing, refer to the "Instruction Manual (Software Vol.3)" attached to the device.

scan parameter

Cine MRI	1.5T	3T
Scan Time	Around 0:14 (Depends on HR and Num. of slice)	Around 0:14 (Depends on HR and Num. of slice)
Sequence	SSFP2D	SSFP2D
TR	4.2	3.2
TE	2.1	1.6
Plane	Arbitrary	Arbitrary
Encode Direction	Direction without wrapping	Direction without wrapping
Num. of slice	Required	Required
Slice thickness	8.0/0.0	8.0/0.0
Matrix	192×208	192×208
FOV	36×36	36×36
No PE Wrap	1.0/1.0	1.0/1.0
SPEEDER	On/2.0	On/2.0
NAQ	1.0	1.0
Flip Angle	90（maximum）	60（maximun）
FatSAT	Off	Off
Num. of Segment/Type	12/ViewSharing	12/ViewSharing
Gating Source	ECG	ECG
Gating Mode	Retrospective	Retrospective
%R-R	120	120
Delay Time	0	0
Num. of Phase	maximum	maximum

Provided by "Canon Medical Systems MRI application support service"